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Intro

This document outlines the technical specification and API design for a Software as a Service (SaaS) platform targeting small and medium-sized businesses in the catering (bars, cafes, restaurants) and beauty (barbers, hairdressers, spas) sectors.

The primary goal of this project is to create a detailed and implementable technical blueprint. This document will be used by a separate development team to build the final product. Therefore, clarity, precision, and completeness are of utmost importance. The project simulates a real-world scenario where the design and implementation phases are handled by different teams, highlighting the critical role of comprehensive documentation.

The system is designed to be operated by business employees, with customers primarily interacting during payment and booking. The core functionalities are divided into three main areas :

- **Order Management:** *Creating, modifying, canceling, and processing payments for customer orders. This includes features like split checks, tips, and discounts.*
- **Service Reservations:** *Booking, modifying, and canceling appointments for services.*
- **System Management:** *Tools for managing products, services, taxes, users, and merchant details.*

This specification will provide business flows, high-level architecture diagrams, a detailed data model, and API contracts using the OpenAPI standard.

Business Flows

Order Flows

Create

Preconditions:

Flow

1. Clients arrive, take a seat, and look over the menu.
2. The employee greets them, opens the order notebook, assigns a new order number and informs them about unavailable products.
3. The client states the first item, including chosen ingredients/options and the quantity.
4. The employee notes any extra-cost ingredients or add-ons.
5. If the client agrees, the employee records the item and its quantity.
6. The employee asks whether the client would like to add more items.
7. Repeat steps 3–6 until the client is finished, making any edits as requested.
8. The employee totals the order, factoring in any item-level discounts, serving fees, and taxes and writes down the subtotal.
9. The employee may apply an additional discount to specific items or to the whole order –if the necessary permission is given or ordered by the manager–, then records the final total.
10. When ready, the client decides whether to split the bill among members; if split, each member selects a payment method for their share and may add a tip individually.
11. Pays the order by the selected payment method.
12. The employee collects the payment, and marks down the order is paid/closed.

Tables/Entities

- | | |
|------------------|-----------------------|
| • Employee | • Product |
| • Order | • Product Ingredients |
| • Food Catalogue | • Discount |
| • Cart | • Payment |

- Tip
- Stock

Components

- User Manager/Permissions
- Order Management
- Payment Management
- Inventory

Actions

- Client tips.
 - Client edits the order before being closed/paid.
 - Client selects payment method and pays.
 - Employee applies discounts
 - Employee checks about unavailable items.
-

Edit

Preconditions:

- An order has already been created and is still open (not paid yet)
- The employee is authorized to make changes

Flow:

1. The customer requests for a change to his order
2. The employee retrieves the open order details
3. The employee adjusts the order as requested:
 - Adding new items
 - Removing items no longer wanted
 - Changing quantities or options
4. The employee confirms the changes with the customer
5. The updated order continues as normal until payment

Tables/Entities

Components

Actions:

- The employee reviews the customers requests for changes
 - The employee adjusts the order by adding,removing, or updating items
-

Cancel

Preconditions:

- An order has been created but is still **open** (unpaid/open).
- The customer requests to cancel the order, or the employee decides to cancel it due to a mistake or unavailability.
- The employee has authorization to cancel orders.

Flow:

1. The customer informs the employee that they no longer want the order, or the employee identifies that the order should not continue (e.g., wrong entry, unavailable items, customer leaves).
2. The employee reviews the open order.
3. The employee confirms the cancellation with the customer if it is possible.

Tables/Entities

Components

Actions

- Employee cancels an order.
 - Employee confirms the cancellation.
-

Refund

Preconditions:

- The order has already been closed and paid
- The customer requests a refund

Flow:

1. The customer explains the reason for the refund.
2. The Manager reviews the request.
3. If the refund is approved, the employee decides whether it will be a full or a partial refund
4. The employee returns the agreed amount the the customer using the appropriate method(cash,card,gift-card)
5. The employee gives the customer a refund receipt or note of confirmation
6. The order status changes to refunded.

Tables/Entities

- Refunds

Components

Actions

- The Manager decides whether the refund will be full or partial.
- The employee refunds.

Reservation Flows

Creation

Preconditions

- The business accepts reservations (services).

- Customers can request reservations in advance (by phone, in person, or other accepted means).
- Staff are authorized to accept and confirm reservations.
- Availability information (tables, employees, timeslots) is up to date in the system.

Flow:

1. A customer calls or arrives to request a reservation.
2. Customers may request a service with a specific employee at a specific date and time. If no specific employee is requested one will be assigned randomly to the reservation.
3. If a slot is free, the staff member pencils in the customer's name and details.
4. The staff member assigns an employee to the reservation by writing their name next to the entry.
5. Confirmation is given verbally or by handing the customer a small confirmation slip.

Table/Entities

- Customer – Stores customer details such as name, contact information, and preferences.
- Reservation – Holds details of the booking, including date, time, number of guests, status, and deposit information.
- Schedule
- Employee – Represents staff members who can be assigned to reservations.
- Service
- Payment/Deposit – Records prepayment or deposit transactions linked to a reservation.
- Notification – Stores confirmation or update messages sent to customers.

Components

- Reservation Management System: Handles reservation creation, availability checking, and employee assignment.
- Payment System: Processes deposits or prepayments.
- Notification Service: Sends confirmation messages (SMS, email).

Actions

- The customer performs a reservation on a free timeslot.
 - An employee is assigned to a reservation.
 - Confirmation is communicated to the customer.
-

Edit

Preconditions

- An active reservation already exists.
- The person requesting the change (customer or staff) is authorized to do so.

Flow

1. The customer requests a change to an existing reservation (such as time, date, or service).
2. The staff member reviews the current reservation details. Availability is checked for the new requested conditions.
3. If suitable capacity is available, the reservation is updated by the employee.
4. The customer is informed of the changes and any implications (e.g., new time, pricing, or special arrangements).

Tables/Entities

Components

Actions

- Reservation is edited.
- The customer receives confirmation of the modification.

Cancel

Preconditions

- A reservation exists but has not yet been fulfilled.
- The customer or business decides to cancel it.

Flow

1. The customer or staff member initiates a cancellation request.
2. The existing reservation details are reviewed.
3. If the cancellation is within the permitted time frame, it is approved.
4. The reservation is marked as canceled.
5. If a deposit or prepayment was made, applicable refund or forfeiture rules are followed.
6. The customer is informed that the reservation has been canceled and, if relevant, of the refund or penalty conditions.

Tables/Entities

- Cancellation fees

Components

Actions

- Customer cancels a reservation.
- Customer receives confirmation of cancellation.
- Cancellation fees apply

Tax Management

Create (Tax Setup and Configuration)

Preconditions

- The business operates in a jurisdiction where taxes (e.g., VAT, sales tax, service tax) are applicable.
- The organization has defined official tax policies and approved calculation rules.

- Authorized personnel (e.g., manager, administrator) are allowed to create and configure tax rules.

Flow

1. The Owner or authorized administrator identifies the need to create a new tax rule or policy (e.g., new VAT rate or regional tax).
2. The manager-owner defines:
 - Tax name/type
 - Rate/percentage (e.g., 21%).
 - Description
 - Effective Date

Tables / Entities

- Tax

Components

- Tax Management/Configuration Module

Actions

- A new tax is created by an authorized personnel
-

Edit (Adjustment)

Preconditions.

- A change (e.g., rate adjustment or policy update) has been authorized by management.
- The individual performing the edit has the necessary administrative permissions.

Flow

1. A admin or owner decides that an existing tax rule needs to be updated.
2. The proposed changes—such as a new rate or a modified scope—are written on a **change request form**.
3. This form is reviewed and signed by an authorized person (e.g., the business owner or an accountant).

4. The admin informs all relevant staff (e.g., cashiers, accountants) of the change and its effective date.

Tables / Entities

Components

- Tax Configuration Module
- Change Management Register
- Audit Log System

Actions

- Manager reviews and updates existing tax rules.
 - Organization approves and applies the change.
 - System enforces new tax rates automatically.
 - Updates are logged for traceability.
-

Delete (Deactivate or Remove Tax Rule)

Preconditions

- The tax rule is no longer valid (e.g., policy repeal, expired promotion, or jurisdiction change).
- Admin has confirmed that the rule is not applicable to any active orders or pending transactions.
- The deletion or deactivation has been formally authorized.

Flow

1. The admin identifies a tax rule that is obsolete or no longer applicable.
2. Before removal, the admin ensures:
 - All transactions using the tax are completed.
 - No pending refunds or orders depend on that tax.
3. The admin updates the system to deactivate the tax rule (recommended) or delete it if permanent removal is allowed.
4. The date and reason for removal are recorded in the system (e.g., "VAT replaced by 2025 regulation").
5. The change is signed off by an authorized individual for accountability.
6. The rule is archived in historical tax records for audit purposes.
7. The system stops applying the tax to new transactions immediately.

Tables / Entities

- Tax
- AuditLog / Archive

Components

- Tax Configuration Module
- Audit Log / Archive System

Actions

- Organization identifies expired or invalid tax rules.
 - Administrator deactivates or deletes the rule.
 - The system removes the rule from future calculations.
 - Record of deletion is logged for compliance.
-

Cancel (Invalid or Rejected Tax Configuration Attempt)

Preconditions

- A manager or employee attempted to create or modify a tax rule.
- The setup request was found to be invalid or unauthorized (e.g., incorrect rate, missing approval, or data error).
- The tax rule has not yet been applied to live transactions.

Flow

1. The manager or administrator initiates a new tax setup or modification request.
2. The organization reviews the details and finds missing or incorrect information (e.g., invalid tax percentage, unapproved exemption).
3. The manager marks the request as canceled or not approved before activation.
4. No tax record is created or, if one was started, it is marked as "Canceled" for reference.
5. The person who submitted the request is informed of the reason for rejection and any corrective actions required.
6. The cancellation is logged in the Tax Setup Review Notes or Management Register.
7. If corrected and reapproved later, a new creation process can begin.

Tables / Entities

- Tax

Components

- Tax Configuration Module

Actions

- Manager reviews tax setup request.
- Invalid configuration is identified and rejected.
- The cancellation reason is documented.
- Requester may reapply after correction.

User Management

Preconditions:

- A new employee has been officially hired by the business.
- A manager or owner has the authority to add new employees to the company records.

Flow

1. A new hire provides their necessary personal and contact details to the manager.
2. The manager creates a new entry for the employee in a physical **Employee Register** or personnel file.
3. The manager assigns a specific role (e.g., "Waiter," "Cashier," "Supervisor") to the employee by writing it in the register.
4. The manager verbally explains the responsibilities, limitations, and permissions associated with the assigned role.
5. The new employee's file is stored securely with other personnel records.

Tables/Entities

- Role
- User
- Permission

Components

- User Registration Module
- User profile Management

- Role Management System
- Access Control System

Actions

- Creating a new user.
 - The organization assigns an appropriate role and permissions.
-

Edit

Preconditions:

- An employee's record already exists in the Employee Register.
- A manager has decided to change the employee's role, permissions, or personal details.

Flow

1. The manager identifies the need to update an employee's record (e.g., due to a promotion, change of address, or new responsibilities).
2. The manager retrieves the employee's existing file from the Employee Register.
3. They cross out the outdated information and clearly write the new details, adding a note with the date and reason for the change.
4. The manager discusses the changes with the employee, particularly if they involve new duties or permissions.
5. The updated file is returned to the secure storage location.

Tables / Entities

Components

Actions

- **Modify Profile:** The system allows an administrator to edit a user's details (e.g., name, contact info).
- **Change Role:** The system enables changing the role assigned to a user, which automatically updates their permissions.

Delete

Preconditions

- An employee is leaving the organization (e.g., resignation, contract termination).
- The employee has completed all assigned duties and there are no remaining obligations.
- The organization authorizes the removal of the employee's name and permissions from the employee register.

Flow

1. The manager retrieves the departing employee's file from the Employee Register.
2. They mark the file clearly as "Inactive" or "Terminated," noting the final date of employment.
3. The manager signs and dates the entry to formalize the record.
4. The physical file is moved from the active employee section to an archive for long-term storage.

Tables / Entities

Actions

- An employee account can be deactivated and deleted.

Discount Management

Notes

Product stands for both food products and services(haircut, massage etc)

Creation

Preconditions

- A responsible employee (e.g., a manager or owner) with the authority to create promotions has been designated.
- TO CHANGE TO TECHNICAL The employee is logged into the system and is authorized with the necessary permissions to create discounts.

Flow

1. The responsible employee decides to introduce a new discount.
2. They determine if the discount applies to a **specific product/service** or an **entire order/reservation**.
3. They decide on the discount format—either a **specific monetary amount** or a **percentage rate**—and write down its value.
4. Optionally, he decides whether a minimum price to apply the discount should exist, and the duration of the discount.
5. They also decide if there should be a **limit on the number of uses**.
6. The employee records these rules in a central logbook and creates a written notice for staff.
7. The new discount is communicated to all staff members, and the notice is posted in a staff-only area.

Tables/Entities

- Discount

Components

- Discount Management

Actions

- Employee selects the discount scope (product-specific or order-wide).
- The employee defines the discount type and value.
- The employee sets the conditions for the discount (duration, minimum price).

Edit

Preconditions

- A promotional offer is currently active.
- A responsible employee (e.g., a manager or owner) with the authority to create promotions has been designated.

Flow

1. The responsible employee decides to alter an existing discount (e.g., change the percentage, extend the duration).

2. The responsible employee finds the original written notice or logbook entry for that discount. They cross out the old information and clearly write the new terms next to it, adding the date of the change.
3. The responsible employee verbally informs all staff members of the update to ensure everyone is aware of the change immediately.
4. The updated notice is posted for staff to reference.

Tables/Entities

Components

Actions

- The employee modifies the discount details on the physical notice/logbook.

Delete

Preconditions

- An existing discount needs to be discontinued.
- A responsible employee (e.g., a manager or owner) with the authority to create promotions has been designated.

Flow

1. The responsible employee communicates to all employees that the discount is no longer valid as of a specific date and time.
2. The responsible employee removes the physical notice about the discount from the staff area.
3. In the company logbook, the responsible employee marks the discount as "expired" or "discontinued" and notes the date it ended.
4. Employees stop applying the discount to all new customer orders.

Actions

- Termination of a discount.

Technical flows and UI Wireframes

This section covers parts that the system will automate/digitalize.

Order

New Order

Order #1024

Some items are currently unavailable

Starters

Mains

Desserts

Drinks

Caesar Salad

\$8.99

+Add

Garlic Bread

\$5.50

+Add

Onion Rings

\$6.25

+Add

Bruschetta

\$7.75

+Add

Order Summary

Grilled Salmon

+ Extra Sauce

\$18.99

Margherita Pizza

+ Extra Cheese, + Jalapeños

\$15.50

Chicken Burger

No add-ons

\$12.75

Special Requests

Subtotal:

\$47.24

Discount:

-\$4.72

Total:

\$42.52

Apply Discount

Split Bill

Proceed to Payment

Cancel Order

figure x: Order Overview and product selection



Cappuccino

Rich espresso with steamed milk and foam

\$4.50 - 1 +

Size

Small

Medium

Large

Sugar

None

Low

Medium

High

Extras

Whipped Cream \$0.50 +

Chocolate Syrup \$0.75 +

Caramel Drizzle \$0.75 +

Order Summary

Cappuccino (Medium)	\$4.50
Whipped Cream	\$0.50
Total	\$5.00

Cancel

Add to Cart

Figure x: Selecting product



ingredients(all categories have been extended for showcase reasons)

Technicalities

Preconditions:

- We consider that there are no unavailable items in this case.
- The employee is authorized to implement additional discounts.
- Select product may contain search using categories or a search bar

Explanations

The "Retrieve Shop Menu" action initiates a real-time data query to the system's database to fetch the complete and most current set of all active business rules. This ensures that any newly created or modified order discounts, product-specific discounts, and tax regulations stored in the management modules are immediately accounted for.

The price is not saved, as it's calculated on the fly by summing each product's base price + price of extra ingredients, any product discount and tax of the product.

Products that do not support ingredients won't have a product overview.

Products may have default/preselected ingredients that will be automatically selected as defaults. Employee may choose to unselected, add additional, or change ingredients.

Payment

UI wireframes

Order Summary
Order Number: {ord}

Your bill
Amount {totalAmount} €

Left to pay {leftToPay} €

Order Cart

{Quantity}	{productName} Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non	Line item	{productTotalprice} € Discount: {p_dis} € Tax: {p_tax} €
{Quantity}	{productName} Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non	Line item	{productTotalprice} € Discount: {p_dis} € Tax: {p_tax} €

Split

Pay all

Select Items to Pay

Unpaid items

{selectedQuantity}

{productName}

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non

{remainingQuantity}

{productName}

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non

{selectedByOtherQuantity}

{productName}

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non

Cart

{productTotalPrice} €

Discount: {dis} €

Tax: {p_tax} €

{productTotalPrice} €

Discount: {dis} €

Tax: {p_tax} €

Line Item

{productTotalPrice} €

Discount: {dis} €

Tax: {p_tax} €

Paid Items

{paidQuantity}

{productName}

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non

{productTotalPrice} €

Discount: {dis} €

Tax: {p_tax} €

Items Paid by:

Your Name

Add tip:

Tip amount: 0.005

Pay

This is the wireframe when the payment is split. The different colored quantity chips mean

- Blue — “Selected by you”
Shows the quantity you’re about to pay for on this item. Click to change the quantity.
- Black — “Available”
Quantity that is unselected and still available for anyone at the table to pick.
- Gray — “Locked by others”
Quantity temporarily unavailable because another guest has selected it. It becomes available again if they reduce their selection or cancel before paying.
- Green — “Paid” (in the Paid items section). Quantity that has already been paid and is no longer editable.

How selection works:

1. Click a chip to allocate or adjust the quantity you intend to pay.
2. If you set the quantity to 0, the item returns to the Unselected (black) state.
3. Items move to Paid items after a successful payment.
4. Gray (locked) quantities automatically unlock if the other guest cancels or completes payment.

Payments Overview

Order Number: {ord}

Your bill

Amount

Payment #1

By : {usernamePaidOrder}

Product Paid: {productPaidInThisPayment}

{paymentAmountPaid}€

Payment #2

By : {usernamePaidOrder}

Product Paid: {productPaidInThisPayment}

{paymentAmountPaid}€

Payment #3

By : {usernamePaidOrder}

Product Paid: {productPaidInThisPayment}

{paymentAmountPaid}€

Payment #4

By : {usernamePaidOrder}

Product Paid: {productPaidInThisPayment}

{paymentAmountPaid}€

Left to pay

{leftToPay}

This wireframe represents the payment overview of split payment.

Secure Payment

Card Number

{cardNumber}

Expiry Date

MM/YY

CVC

123

Cardholder Name

John Doe

Subtotal

{subtotalPrice}\$

Discount

{totalDiscounts}\$

Tax

{totalTax}\$

Total

{totalPaymentAmount}\$

{totalPaymentAmount}\$

Your payment details are securely encrypted

stripe

Powered by Stripe

visa

apple

card

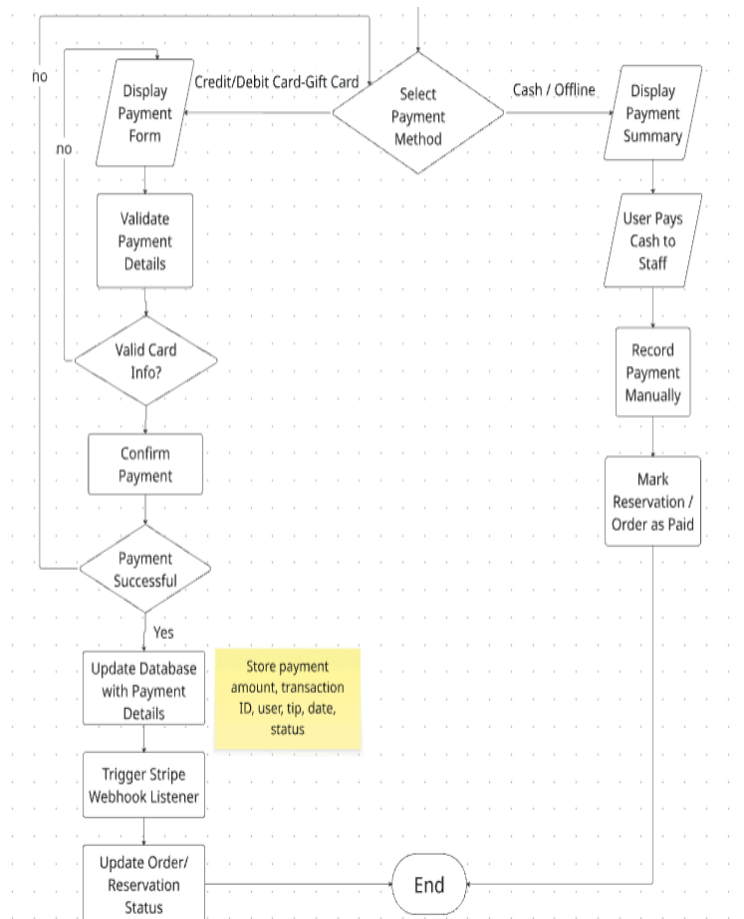
Preconditions

- The order or reservation has been created and is ready for payment.
- The employee handling the payment is authorized to process payments.

Explanations:

- **Price calculation:** The total is calculated on-the-fly as the sum of:
 - **Base price of each product**
 - **Extra ingredient/add-on costs**
 - **Applied product-level or order-level discounts**
 - **Applicable taxes**
- **Split check option:** If the customer wants to split the bill among multiple people, the system divides the total among participants. Each participant can:
 - Select their preferred payment method (cash, card, gift card)
 - Add an optional tip to their choice
- **Payment method selection:** The customer chooses a payment method (cash, card, gift card). If the bill is split, each participant can select their method independently.
- **Transaction recording:** Once payment is received, the system logs the transaction and updates the order status to **paid/closed**.

Payment flow chart



Logging

Preconditions:

We have:

- A consistent logging format
- Secure infrastructure connections
- Proper role-based access and encryption
- Storage capacity and alerting in place
- Clear ownership and compliance agreements

1. Where logs come from

- **Apps & Services:** Your backend, frontend, and APIs.
- **Infrastructure:** Servers, containers, load balancers.

- **Security tools:** Authentication systems, firewalls, etc.

All components write logs in a **consistent JSON format** (with timestamp, service name, environment, severity, and trace ID).

2. Collecting logs

- A **log agent** (like Fluent Bit, Filebeat, or OpenTelemetry Collector) runs on each machine or container.
- It reads local logs and sends them to a **central logging service**.
- Logs are cleaned and filtered — no secrets, passwords, or tokens should appear.

Permissions:

- Agents can only read logs and send them.

3. Ingestion & Storage

- Logs first go into a **queue or stream** so nothing gets lost.
- Then they're sent to a **storage system** for indexing and searching.
- Old logs are **archived or deleted** after the retention period.

Permissions:

- Only system components can write.
- Only specific roles (Admin, SuperAdmin) can delete or change storage settings.

4. Viewing & Using Logs

- Developers and SREs use tools to view and search logs.
- Dashboards show service health, errors, and performance.
- Alerts notify teams when error rates spike or suspicious activity occurs.

Permissions:

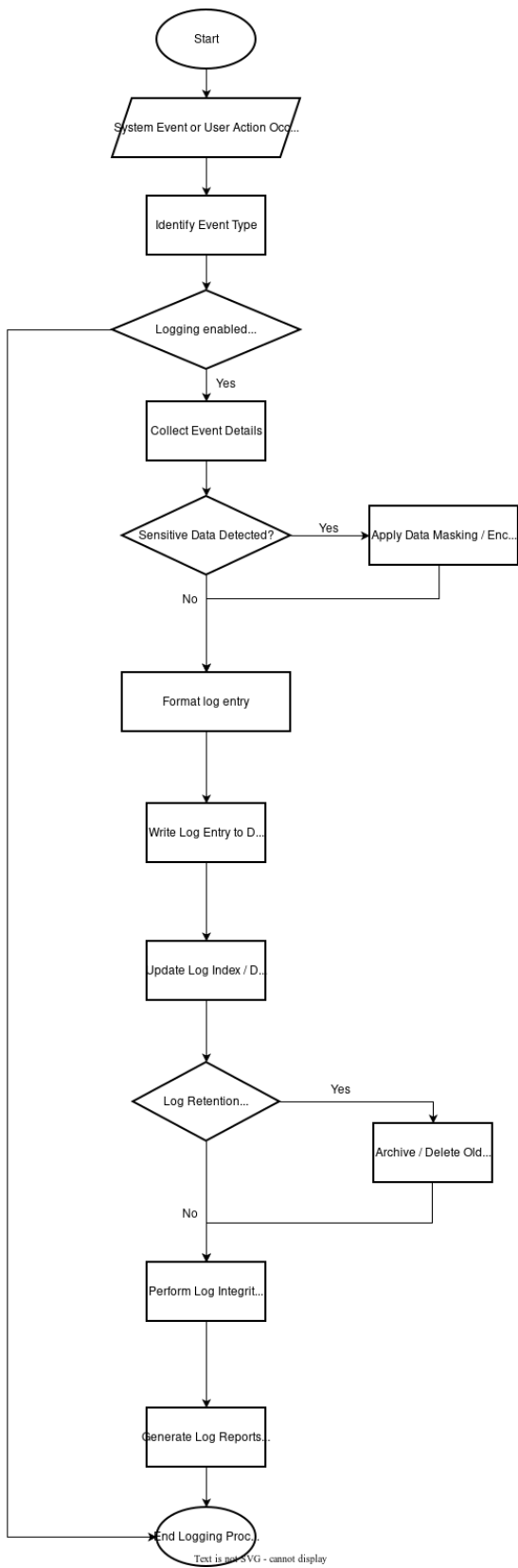
- **Developers:** Read logs for their own services.
- **SuperAdmins:** Full access (read/write/manage).
- **Admins:** Read all logs for investigations.

5. Governance & Safety

- No sensitive data in logs.
- Logs are encrypted (in transit and at rest).

- Each log includes enough info to trace a request end-to-end.
- Access is monitored and audited.

Representation of the flow using a diagram:



New Appointment

1

Customer Info

2

Service

3

Employee

4

Date & Time

5

Review

Customer Information

Full Name

Enter customer name

Phone Number

Email Address

Enter email address

Select Service

Haircut & Styling

60 minutes

\$45

Hair Coloring

120 minutes

\$85

Consultation

30 minutes

\$25

Select Employee

John Smith

Stylist

Jane Doe

Color Specialist

Mike Johnson

Senior Stylist

Any Available

System will assign

Select Date & Time

S M T W T F S

28 29 30 31 1 2 3

4 5 6 7 8 9 10

11 12 13 14 15 16 17

18 19 20 21 22 23 24

25 26 27 28 29 30 1

Available Times for June 14, 2023

9:00 AM

10:00 AM

11:00 AM

1:00 PM

2:00 PM

3:00 PM

4:00 PM

5:00 PM

Review Appointment

Customer

Sarah Johnson

+1 (555) 123-4567

sarah@example.com

Service

Hair Coloring

120 minutes - \$85

Employee

Jane Doe

Date & Time

June 14, 2023

10:00 AM - 12:00 PM

Back

Confirm Booking

Reservation

The creation of a reservation for a service breaks down into four distinct steps.

1. Customer info:

All the info of the customer is given, which include name, phone and email.

2. Employee and service selection: The employee selects the chosen employee from the customer that will perform the service.

3. Date & Time selection: Unavailable and past dates will be marked with grey color. Available date & time will be based on the selected employee's availability.

4. Details review

Technicalities:

Preconditions:

- One phone number country.

1. Upon clicking

create a new reservation the system requests the latest updates regarding timeslots and availability.

- Employee inputs customer data.
- The data is validated.

4. Upon selecting service and employee, the date & time selection updates, holding the availability of the specific employee.
5. Upon clicking confirming booking, the reservation is stored in the database and added to the overview.

figure: Reservation creation Wireframe

Reservations					
STATUS	CUSTOMER	SERVICE	EMPLOYEE	DATE & TIME	ACTIONS
Pending	 Robert Johnson (555) 123-4567	Haircut & Styling 45 min - \$35	 Sarah Miller	Oct 15, 2023 10:30 AM	 
Active	 Emily Davis (555) 987-6543	Manicure & Pedicure 60 min - \$50	 Michael Chen	Oct 15, 2023 11:00 AM	 
Completed	 James Wilson (555) 456-7890	Facial Treatment 90 min - \$80	 Lisa Anderson	Oct 14, 2023 2:15 PM	 
Cancelled	 Patricia Brown (555) 234-5678	Massage Therapy 60 min - \$70	 David Thompson	Oct 13, 2023 4:00 PM	 
Pending	 Thomas Garcia (555) 876-5432	Hair Coloring 120 min - \$95	 Sarah Miller	Oct 16, 2023 9:00 AM	 
Showing 1to 5 of 24 results				Previous Next	

figure: Reservation Dashboard

For edit or deletion operations, a specific reservation would be selected from the reservation dashboard and the wireframe on figure x, would be displayed.

When clicking edit, the same wireframe of that of creation would be displayed and the employee could change what ever necessary.

Appointment Details

X

Pending

Customer

Sarah Johnson

Contact

+1 (555) 123-4567

sarah@example.com

Service

Haircut & Styling

60 minutes - \$45

Employee

John Smith

Date & Time

June 14, 2023

10:00 AM - 11:00 AM

Notes

Customer requested specific stylist

Modify

Cancel

Figure x: Overview of a specific reservation

For edit or deletion operations, a specific reservation would be selected from the reservation dashboard and the wireframe on figure x, would be displayed.

When clicking edit, the same wireframe of that of creation would be displayed and the employee could change what ever necessary. Cancellation can be achieved by clicking the cancel button.

Discount

Discounts

+ Create Discount

Discount Type

Status

Date Range

Search

Start Date

End Date

Q

Search discounts...

DISCOUNT	TYPE	VALUE	TIME PERIOD	STATUS	ACTIONS
Summer Sale 2023	Order Discount	15% off	Jun 1, 2023 - Aug 31, 2023 10 days remaining	Active	
New Customer Discount	Product Discount	\$10 off	No expiration Perpetual	Scheduled	
Flash Sale	Order Discount	50% off	Jul 15, 2023 - Jul 16, 2023 Expired	Inactive	
Product Bundle Discount	Product Discount	25% off	Aug 1, 2023 - Sep 1, 2023 Active	Active	

Showing 1 to 4 of 4 results

Previous

1

Next

Figure x: Discount Management Overview

Create New Discount/Edit Discount

Discount Name
Enter discount name

Products (for product discounts)
Select products to apply discount

Discount Type
☒ Order Discount
 ☐ Product Discount

Value
 Amount Percent

Minimum Order Value
\$0.00

Time Period
 Start Date End Date

☒ No expiration date

Usage Limits
 Total uses Uses per order/product

Status
Select

Cancel Save Discount

Figure x: Create/Edit Discount Page

Technicalities

Preconditions

- An authorized user is logged into the system and has been granted the necessary permissions to access and manipulate discount-related data.

Flow

1. The user initiates the discount creation process, typically by clicking a "Create Discount" button from the main discount management interface.
2. The system presents a form for the user to define the new discount's parameters.
3. The user populates the required fields, including:
 - **Discount Name:** A unique identifier for the discount.
 - **Discount Type:** The user must select either "Order Discount" or "Product Discount."
 - If **Product Discount** is chosen, the user selects one or more products to which the discount will apply.
 - If **Order Discount** is chosen, the user may specify a **Minimum Order Value**.
 - **Value:** The user defines the discount's magnitude, either as a fixed monetary **Amount** or as a **Percent**.
 - **Time Period:** The user sets a **Start Date** and **End Date** to define the discount's validity period. Alternatively, the user can opt for the discount to have **No expiration date**.
 - **Usage Limits:** The user can optionally set constraints on the discount's use, such as the **Total uses** and the number of **Uses per order/product**.

- **Status:** The user assigns an initial status to the discount (e.g., Active, Scheduled, Inactive).
4. The user confirms the settings by clicking "Save Discount".
 5. The system validates the input data and, upon successful validation, saves the new discount record to the database.
 6. The system redirects the user back to the discount overview screen, where the newly created discount is now listed.

Menu Management

- **Manage Ingredients Category**

Manage Category						
Filter				Sort By		
Search categories...				Name A-Z		X Clear Filters
Name	Description	Ingredients	Total Products	Created At	Updated At	Action
Paprika	Hot or Sweet	2	18	2/1/2024	2/28/2024	 
Rice	Very Healthy	2	12	1/15/2024	2/20/2024	 
Sugar	Very Sweet	2	25	1/10/2024	3/5/2024	 

Manage Category Overview: A list of all existing categories is displayed in a table. The table shows the category's Name, Description, the number of Ingredients it uses, the Total number of Products assigned to it, and its creation and last update dates. The actions present for each category to manage it. A search filter is available to find categories, and the list can be sorted (e.g., by Name A-Z).

- **Add/Edit Category**

×

Add Category

Category Name *

Description

Ingredients

+ Add Ingredient

brown

1

white

2

Cancel

Add

×

Edit Category

Category Name *

Description

Ingredients

+ Add Ingredient

brown

1

white

2

Cancel

Update

Add Category: The user can create a new category. The required fields are the Category Name and its Ingredients. A Description is an optional field. Ingredients can be added dynamically using an "Add Ingredient" button, and a quantity can be specified for each ingredient.

Edit Category: The user can modify an existing category. The form is pre-filled with the current category's data (Name, Description, Ingredients). The ingredients can be edited or removed, and new ones can be added, similar to the "Add Category" functionality.

- **Delete Category**

Are you sure?

Are you sure you want to delete the category "Beverages"? This action cannot be undone.

Cancel

Delete

Delete Category: The user can delete a category. A confirmation message appears, stating the name of the category to be deleted and warning that this action cannot be undone.

Manage Product

+ Add Product

Filter

Search products...

Category

All Categories

Status

All Status

Sort By

Name A-Z

X Clear Filters

Name	Category	Description	Price	Status	Created At	Updated At	Action
Corn Cheese Pizza	Pizza	It contains high calestrol.	175	Active	1/15/2024	2/20/2024	  
Dark Chocolate Hummus	Dessert	Two tablespoons pack in 3 grams of fib...	123	Active	1/10/2024	3/5/2024	  
Masala Dosa	Dosa	South Indian Dish with masala.	95	Inactive	1/25/2024	2/15/2024	  
Plain Dosa	Dosa	South Indian Dish	60	Active	1/20/2024	3/1/2024	  
Sprite	Cold Drink	Product to drink.	25	Active	2/1/2024	2/28/2024	  

Manage Product Overview: A list of all products is displayed in a table. The table includes columns for the product's Name, Category, Description, Price, Status (Active/out of stock/Inactive), and creation/update dates. A set of actions is available for each product to manage it. A filtering system is present, allowing users to filter by Category, Status, and sort the list (e.g., by Name A-Z). A search bar and a "Clear Filters" option are also provided.

Are you sure?

Are you sure you want to delete the product "Corn Cheese Pizza"? This action cannot be undone.

Cancel

Delete

- Delete Product**

Delete Product: The user can remove a product. A confirmation message appears, starting the name of the product to be deleted and warning that this action cannot be undone.

Add Product

Product Name *

orange juice

Tax Category *

Beverages

Subcategory *

Juices

Ingredient Category

Sugar

Ingredient

White

☒ White

☐ Brown

Price *

5

Description

very healthy

Cancel

Add

- **Add Product**

Add product: The user can create a new product. The form requires a Product Name, selection of a Tax Category (i.e. Food,Beverages,Service), a Catalogue category(displayed as subcategory), categories that will be available and the list of Ingredients from each Ingredient Category (presented as selectable checkboxes). The ingredient category here is a dropdown menu that when selected, the ingredient list is updated containing all the ingredients that this category supports. The user selects all the ingredients that will be available for selection from an order perspective. Upon selecting at least one object, the ingredient category is automatically added to the product's list of ingredient category.A Price and an optional Description are also part of the form.

Tax management

Tax is a configurable rate defined in the Tax Management module and automatically applied to orders during on-the-fly price calculation based on product tax category and active status.

It is governed by role-based permissions and auditability so that only authorized users can create, update, or deactivate it

An administrator creates a tax rule by providing name, rate percentage, description, status, in the Tax Configuration module. The record is stored as a taxRate entity and enters the workflow with statuses such as pending or active, with creation and update timestamps captured for traceability.

Each product or service, have a tax rate already applied in their base price.

User Management

Core permission catalog (resource:action)

In this system permissions are tied to a role. Each role has predefined permissions which are listed below. Users may have their permissions modified by upgrading/downgrading their role.

Each permission is described by providing the resource, then the action.

User management / RBAC

- user:create
- user:read
- user:update
- user:delete
- role:read
- role:assign
- permission:read

Merchant & branches

- merchant:read
- merchant:update
- merchant:deactivate
- branch:create
- branch:read
- branch:update
- branch:delete

Products & menu

- product:create
- product:read
- product:update
- product:delete
- productType:read/create/update/delete
- ingredientCategory:create
- ingredientCategory:read
- ingredientCategory:update

- `ingredientCategory:delete`
- `ingredient:create`
- `ingredient:read`
- `ingredient:update`

- `ingredient:delete`
- `catalogue:read`
- `catalogue:upsert` — set stock/status per product (your catalogue tab)

Orders & order items

- `order:create`
- `order:read`
- `order:update`
- `order:cancel`
- `order:close`
- `order:refund`

- `order:receipt`
- `orderItem:add`
- `orderItem:update`
- `orderItem:remove`
- `order:apply-discount` — apply an order-level discount

Payments, splits, tip

- `payment:refund` — refund a payment (distinct from `order:refund` if you want finer control)
- `payments:read`

Discounts & taxes

- `discount:create`
- `discount:read`
- `discount:update`
- `discount:delete`

- `taxRate:create`
- `taxRate:read`
- `taxRate:update`
- `taxRate:delete`

Services & schedules

- `service:create`
- `service:read`
- `service:update`
- `service:delete`

- `schedule:create`
- `schedule:read`
- `schedule:update`
- `schedule:delete`

Reservations, deposits, cancellation fees

- `reservation:create`
- `reservation:read`
- `reservation:update`

- `reservation:cancel`
- `reservationDeposit:create`
- `reservationDeposit:refund`

- `cancellationFee:apply`
- `cancellationFee:waive`

Role bundles (predefined)

Employee

- **Users/RBAC:** `role:read`, `permission:read`
- **Products & catalogue:** `product:read`, `productType:read`, `ingredientCategory:read`, `ingredient:read`, `catalogue:read`
- **Orders:** `order:create`, `order:read`, `order:update`, `orderItem:add`, `orderItem:update`, `orderItem:remove`, `order:cancel` (*open only*), `order:receipt`
- **Payments:** `payment:create`, `payment:read`, `paymentSplit:create`, `paymentSplit:read`, `tip:add`
- **Reservations:** `reservation:create`, `reservation:read`, `reservation:update`, `reservation:cancel`
- **Schedules/Services:** `service:read`, `schedule:read`

Owner / Manager

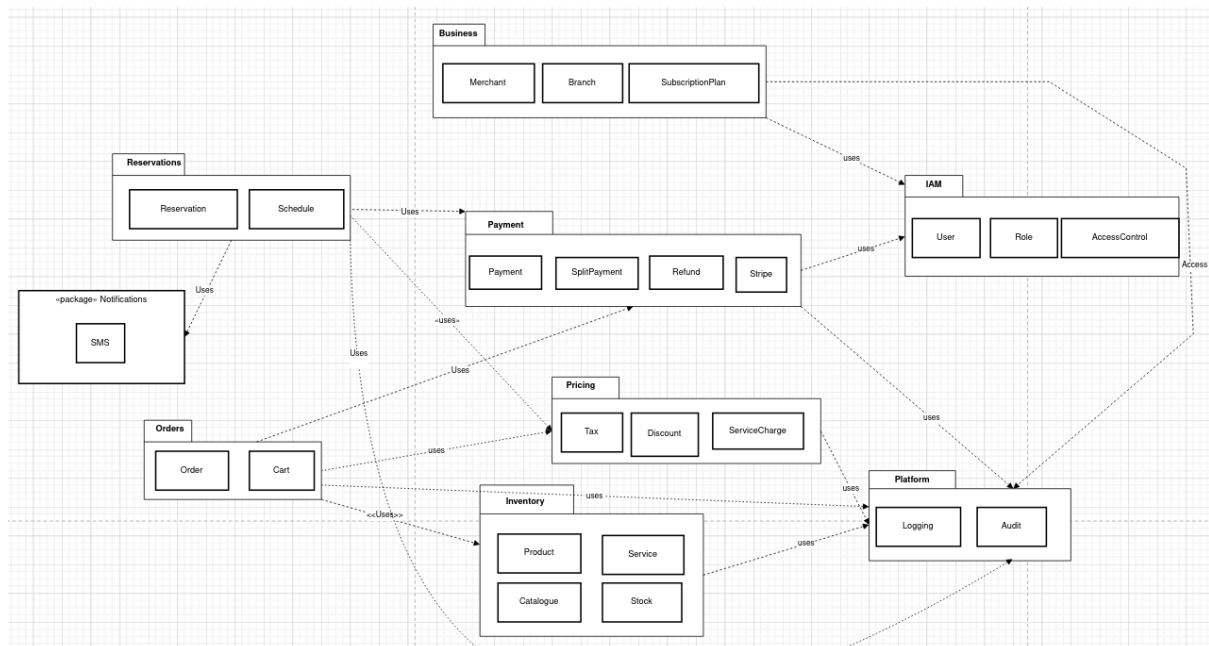
- Everything Employee has, plus:
- **Users/RBAC:** `user:create`, `user:read`, `user:update`, `user:delete`, `role:assign`, `role:read`, `permission:read`
- **Merchant/Branches:** `merchant:read`, `merchant:update`, `branch:create/read/update/delete`
- **Products:** full CRUD on `product`, `ingredientCategory`, `ingredient`, `catalogue:upsert`
- **Discounts & Taxes:** full CRUD (`discount:*`, `taxRate:*`), `order:apply-discount`
- **Orders/Payments:** `order:close`, `payment:refund`, `order:refund`
- **Catalog:** `upsert`
- **Services/Schedules:** full CRUD (`service:*`, `schedule:*`)
- **Reservations extras:** `reservationDeposit:create`, `reservationDeposit:refund`, `cancellationFee:apply`, `cancellationFee:waive`

Guardrail: Owners **cannot** assign the **SuperAdmin** role.

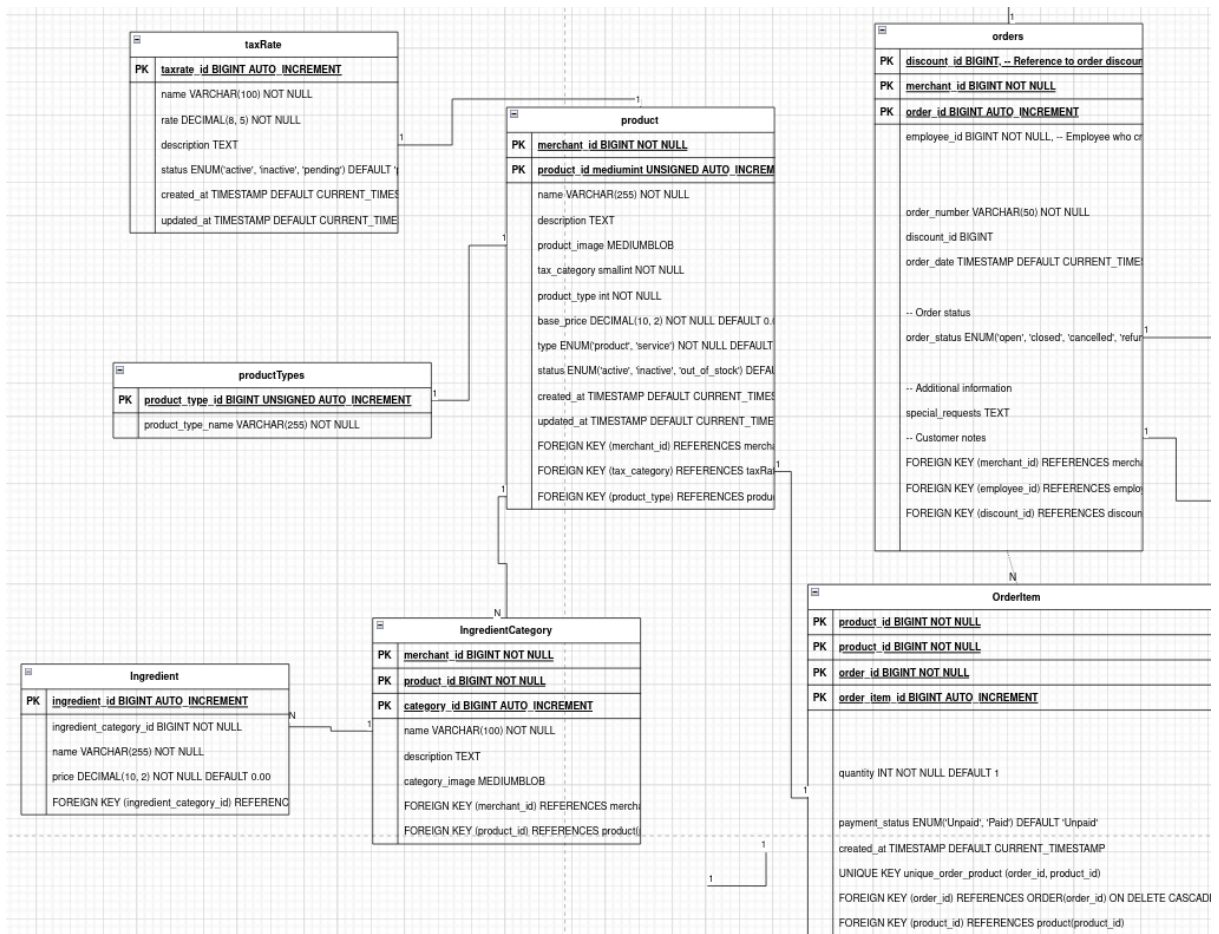
SuperAdmin

- **All of the above across all merchants** (bypass tenant scope).

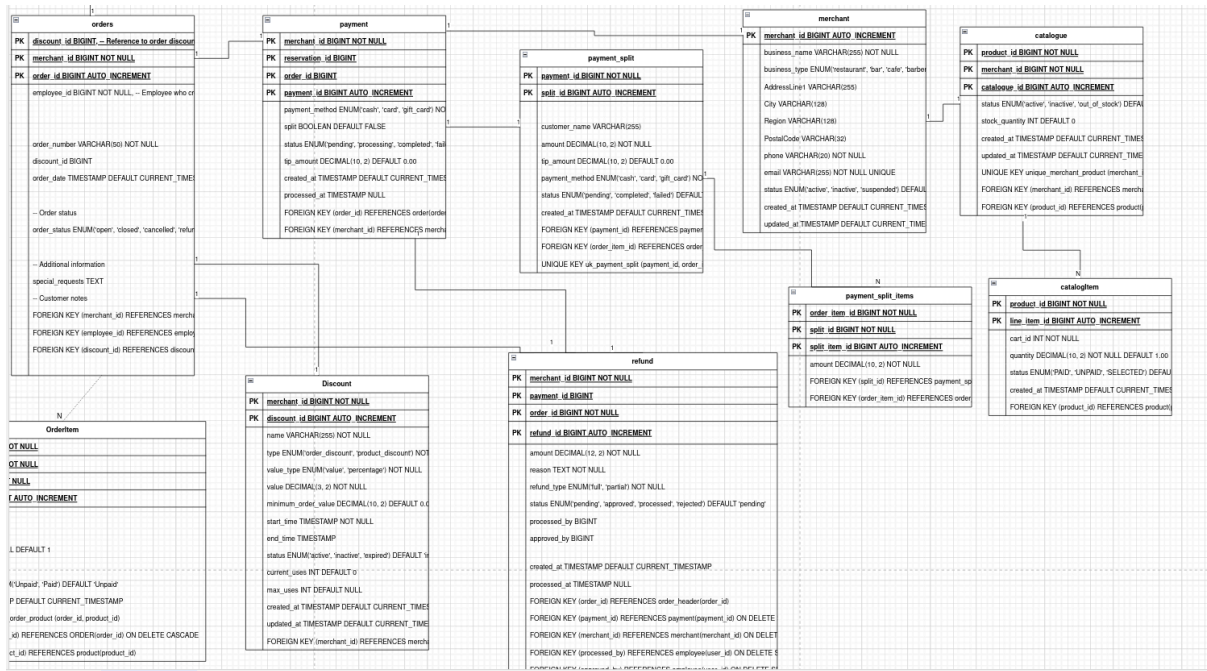
High Level Architecture



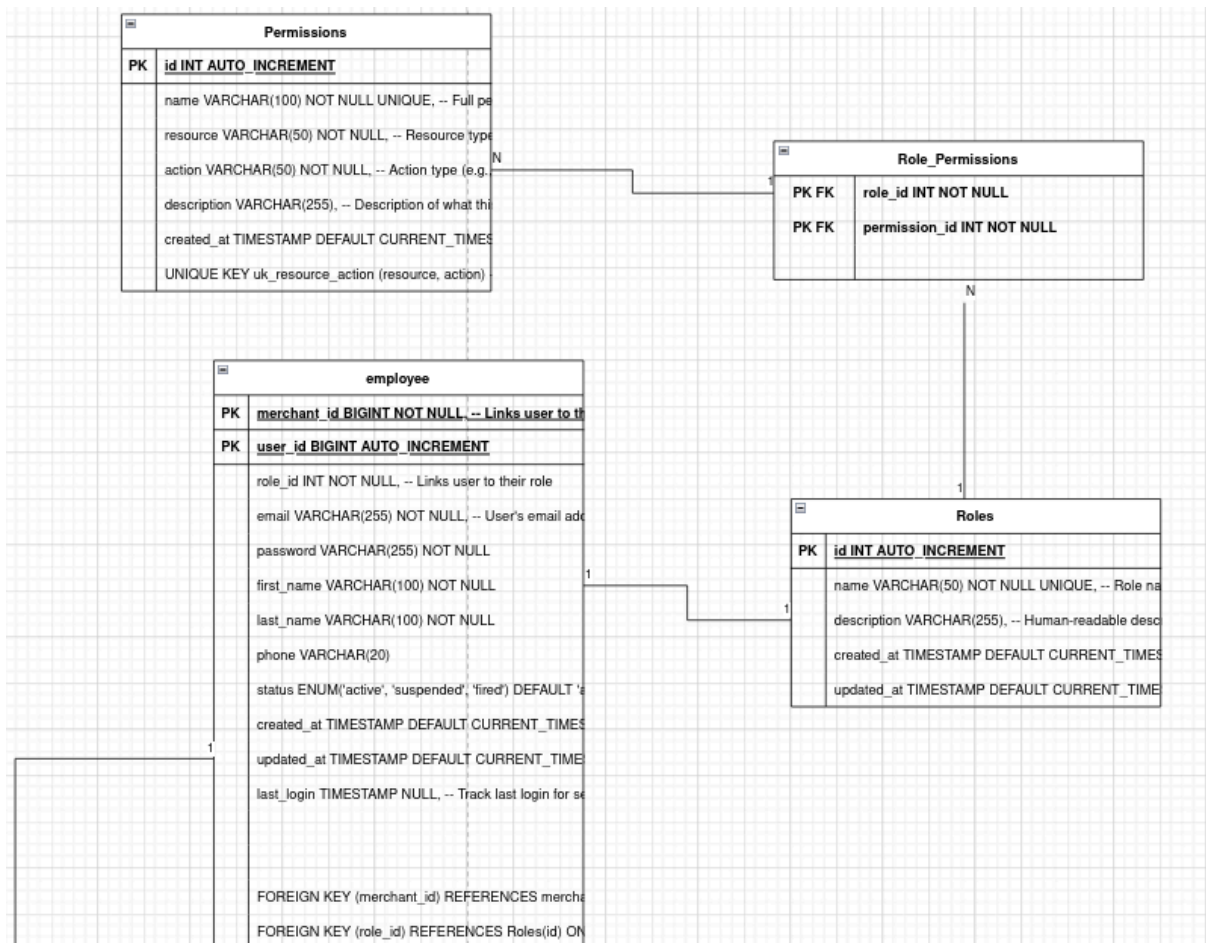
Data Model



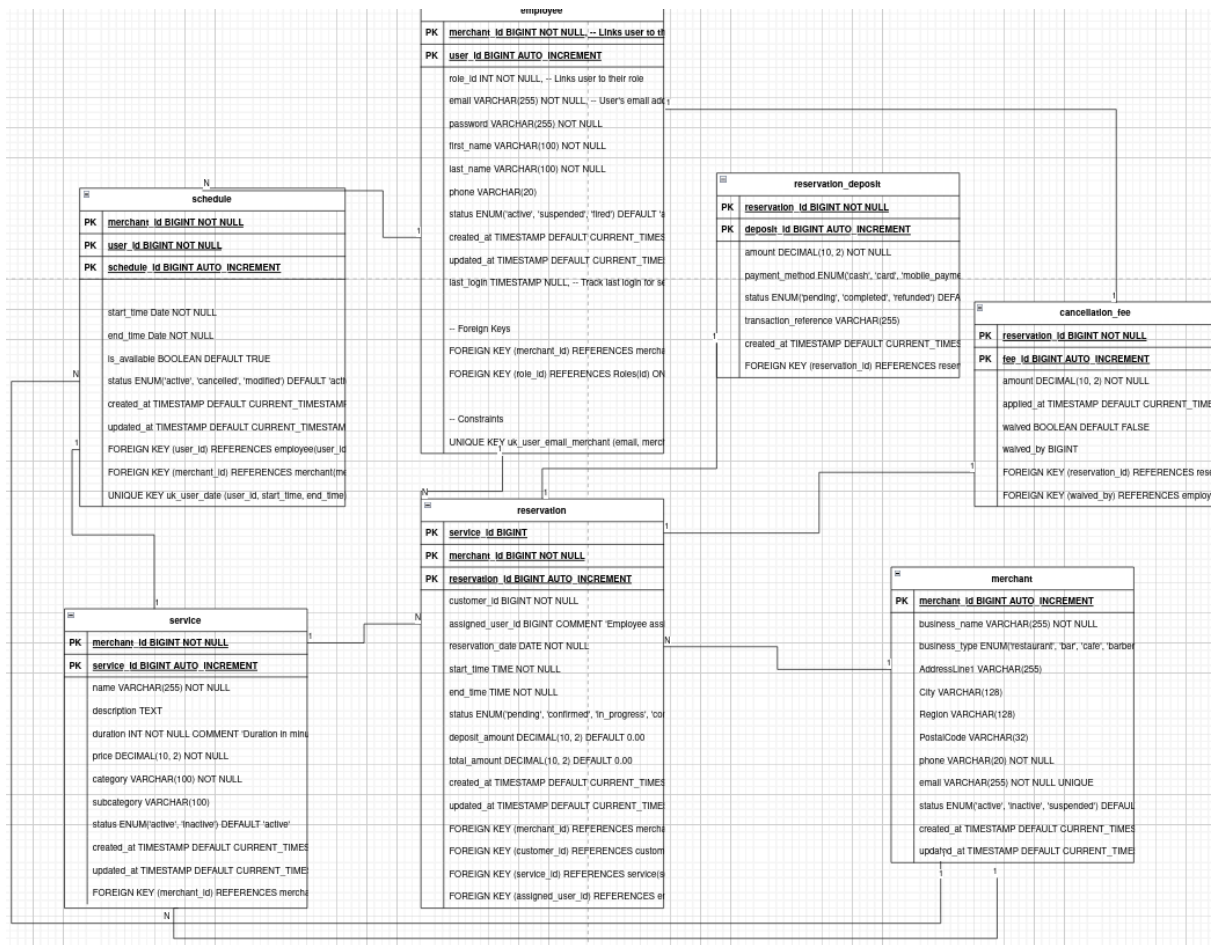
- taxRate-product(1-1),
- productTypes-product(1-1),
- product-IngredientCategory(1-N),
- orders-OrderItem(1-N),
- IngredientCategory-Ingredient(1-N),
- product-OrderItem(1-1)



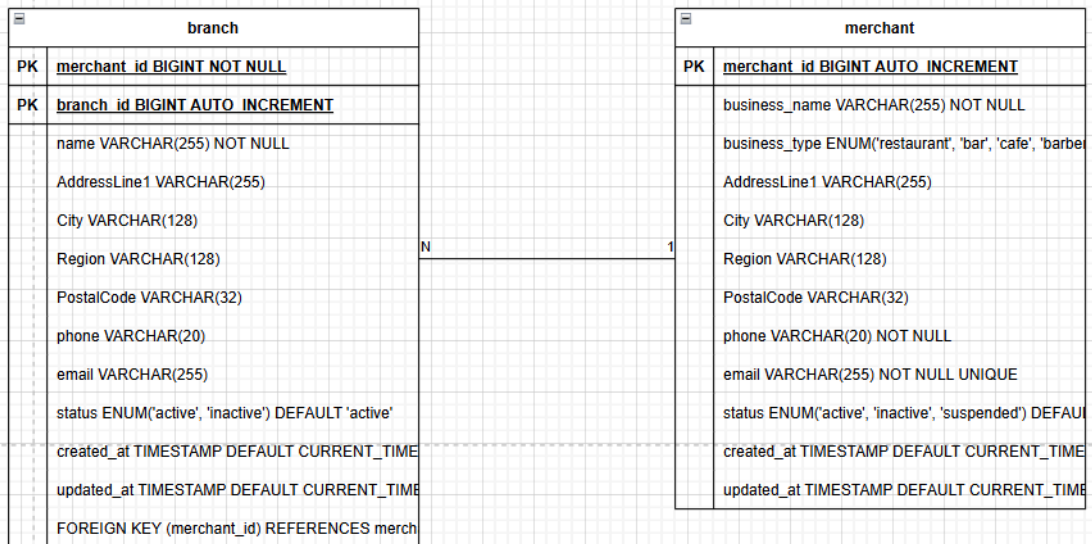
- orders-payment(1-1),
- orders-Discount(1-1),
- orders-refund(1-1),
- payment-payment_split(1-1),
- payment_split-payment_split_items(1-N),
- payment-merchant(1-1),
- merchant-catalogue(1-1),
- catalogue-catalogueitem(1-N),
- payment-refund(1-1)



- Role_Permissions-Permissions(1-N),
- Roles-Role_Permissions(1-N),
- Roles-employees(1-1)



- employee-schedule(1-N),
- employee-cancellation_fee(1-1),
- merchant-schedule(1-N),
- merchant-reservation(1,N),
- service-reservation(1-N),
- service-schedule(1-1),
- reservation-cancellation_fee(1-1),
- reservation-reservation_deposit(1,1),
- merchant-service(1-N),
- employee-reservation(1-N)



- merchant-branch(1-N)